2021-Ph-H1-Q25

Section: Electricity

Topic: Sources, Internal Resistance

A cell with internal resistance 1 Ω is connected to a 5 Ω resistor.

EMF is 6 V. What is terminal voltage?

Solution:

Total resistance = 1 + 5 = 6 Ω

$$I = \frac{E}{R_{\text{total}}} = \frac{6}{6} = 1\,\text{A} V_{\text{terminal}} = IR = 1 \times 5 = 5\,\text{V}$$

Answer: C

Guidance for Students:

Terminal voltage is less than EMF due to internal resistance. Use $V = E - Ir$ or Ohm's Law on external resistor.

Revision Tip:

Terminal voltage = EMF – lost volts:

$$V = E - Ir$$